higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N6

2 AUGUST 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 1.1.1 Resonance on a high voltage power system or damped oscillations due to resonance (1)

1.1.2 Advantage: Used in radio communication circuits

Disadvantage: Causes undesirable voltage and current peaks in power circuits✓ when higher order harmonics are present in the supply (3)

1.2 1.2.1 Overdamping

$$\frac{R^2}{4L^2} \; is\; greater\; than \; \frac{1}{LC}$$

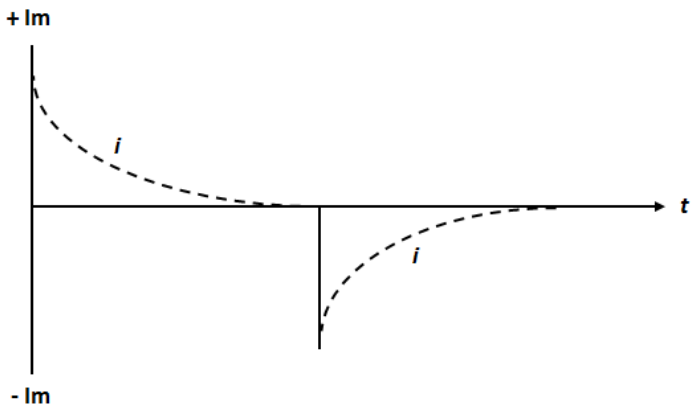
Underdamping

$$\frac{R^2}{4L^2} \; is\; less\; than \; \frac{1}{LC}$$

1.2.2 Overdamping: Sluggish circuit response
Underdamping: Fast circuit response

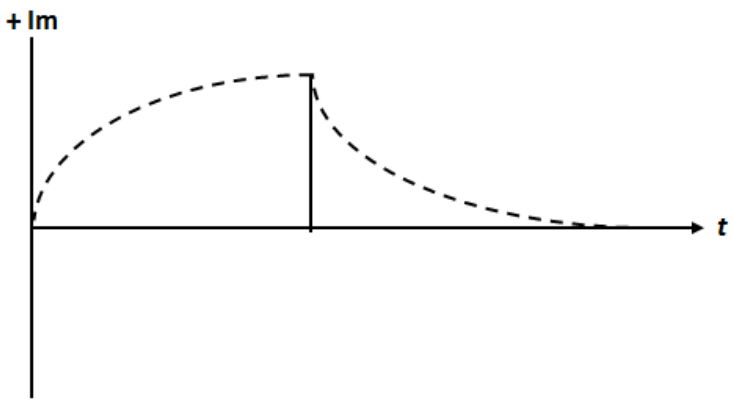
(2 × 2) (4)

1.3



(2)

1.4

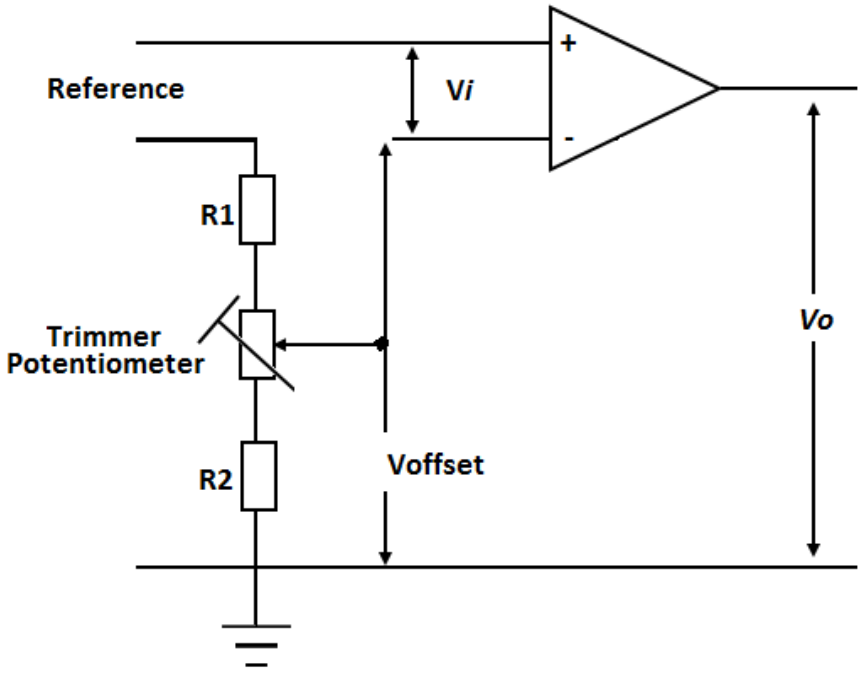


(2)
[12]

QUESTION 2

2.1
- Analogue systems
- Digital systems (2)

2.2



(5)

2.3 Offsetting (1)

2.4 2.4.1 Mark

2.4.2 Space

2.4.3 Mark

2.4.4 Space

(4 × 1) (4)
[12]

QUESTION 3

3.1 Unique ability of certain materials✓ in which the crystal structure undergoes dimensional changes✓ when the materials are subjected to oscillating magnetic or electric fields✓ (3)

3.2
- Magnetostrictive core
- Coupling cone
- Tool tip (3)

3.3
- Permanent magnet
- Feedback winding
- RC tuning circuit (3)

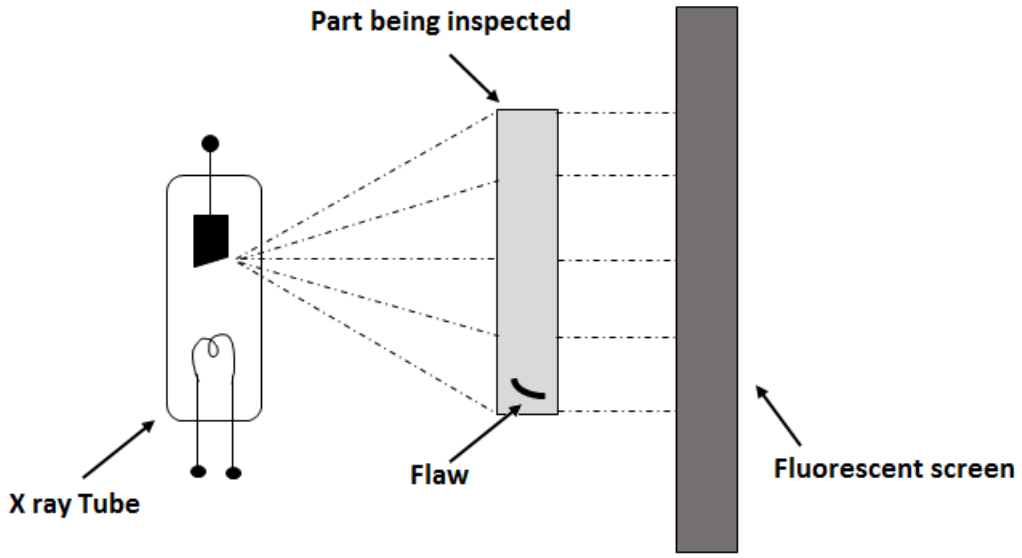
3.4 3.4.1
- It consists of an enclosed tube✓ filled with an inert gas.✓
- Inside the tube there are two electrodes✓ in the form of either rods or circular plates✓ to which a high DC voltage is applied.✓

3.4.2
- There is a high voltage between the electrodes in the gas-filled tube.✓
- If radiation particles enter the gas-filled tube, ionisation✓ between the atoms of the gas inside the tube and the atoms of the radio-active material occurs.✓
- This ionisation results in conduction taking place inside the tube.✓
- The resulting current pulse due to conduction is amplified electronically and used to activate a warning system.✓

(2 × 5) (10)
[19]

QUESTION 4

4.1

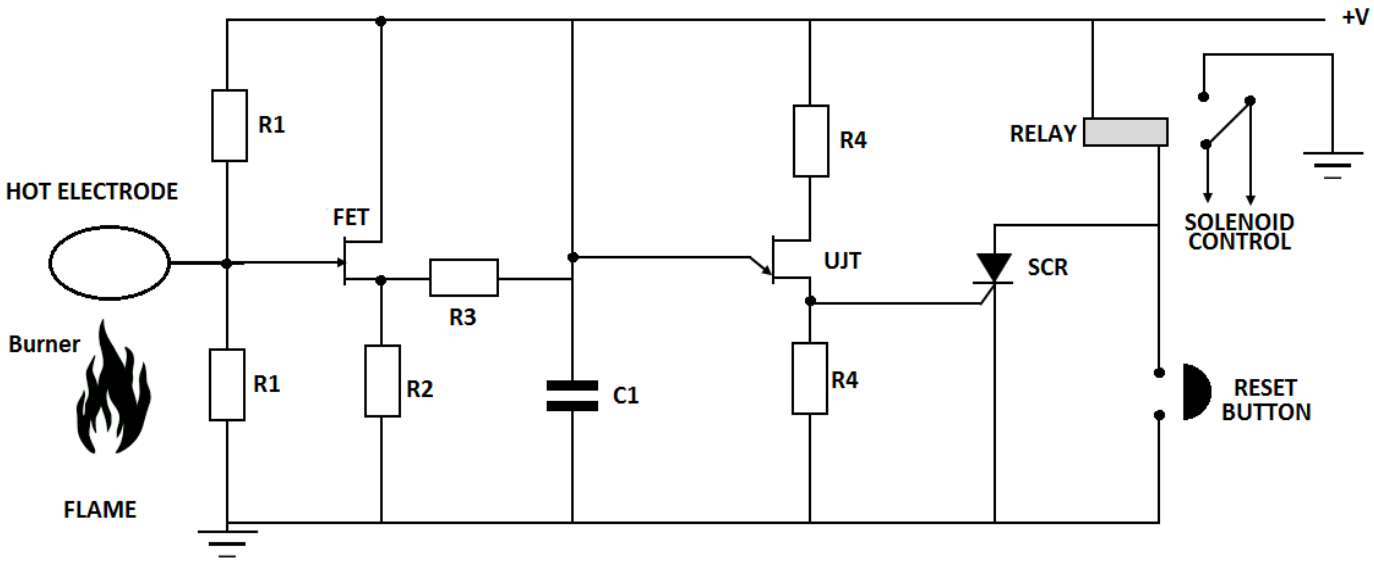


(4)

4.2
- An operator views the image of the casting on a fluorescent screen in a darkened room.✓✓
- If the casting has a flaw or a crack it shows up as a dark line or a shadow✓ on the fluorescent screen.✓ (4)

[8]

QUESTION 5



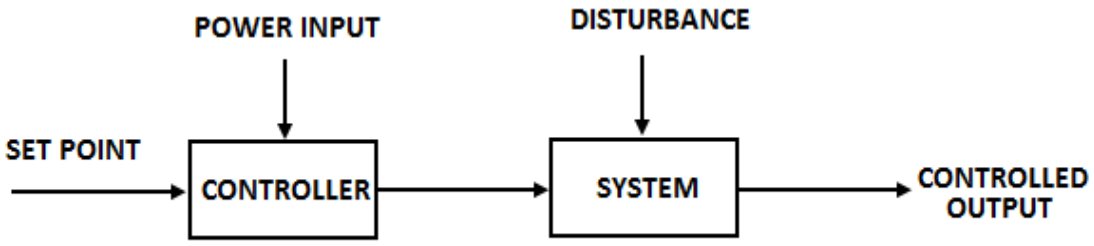
[11]

Mark allocation:

- All resistors, C_1, reset button, hot electrode, flame, burner, relay, +v supply, and ground terminal must be awarded ½ marks.
- FET, UJT, SCR, and solenoid control = 1 full mark each

QUESTION 6

6.1



(4)

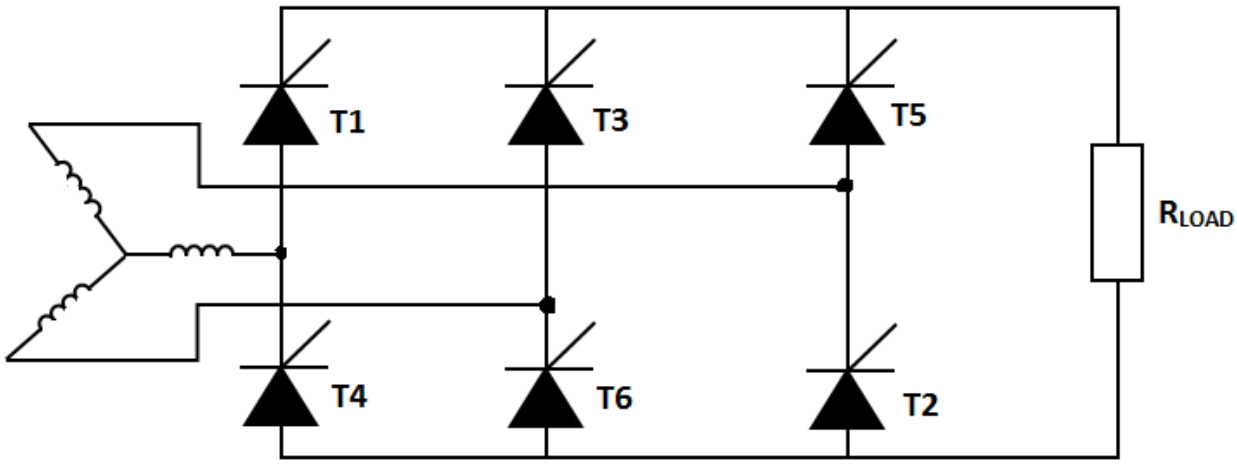
6.2 Computer case VA = 230 × 2 = 460 VA✓

Smart monitor VA = 230 × 1,5 = 345 VA✓
Inkunzi printer = 230 × 0,3 = 68 VA✓
Total VA = 460 + 345 + 69 = 874 VA✓✓
Suitable UPS = 900 VA✓

(6)
[10]

QUESTION 7

7.1



(8)

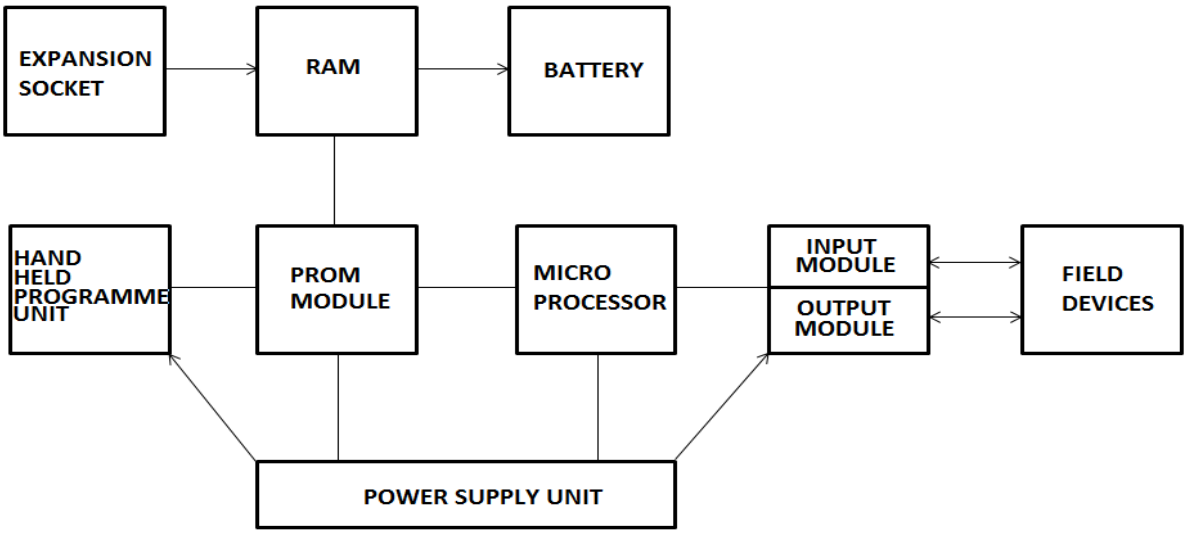
7.2

- Unlimited voltage and current ratings
- Instant turn-on and turn-off times
- Zero leakage current
- Zero conduction and switching losses
- Ability to withstand current overloads and voltage transients
- Zero gate firing power requirement
- Easy to protect against spurious turn-on and fault conditions
- Low cost and ease of assembly (8)

[16]

QUESTION 8

8.1



(8)

8.2
- Electrical engineering
- Instrumentation engineering
- Chemical engineering
- Mechanical engineering (4)

[12]

TOTAL: 100

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